

SMART INDIA HACKATHON 2025



TITLE PAGE

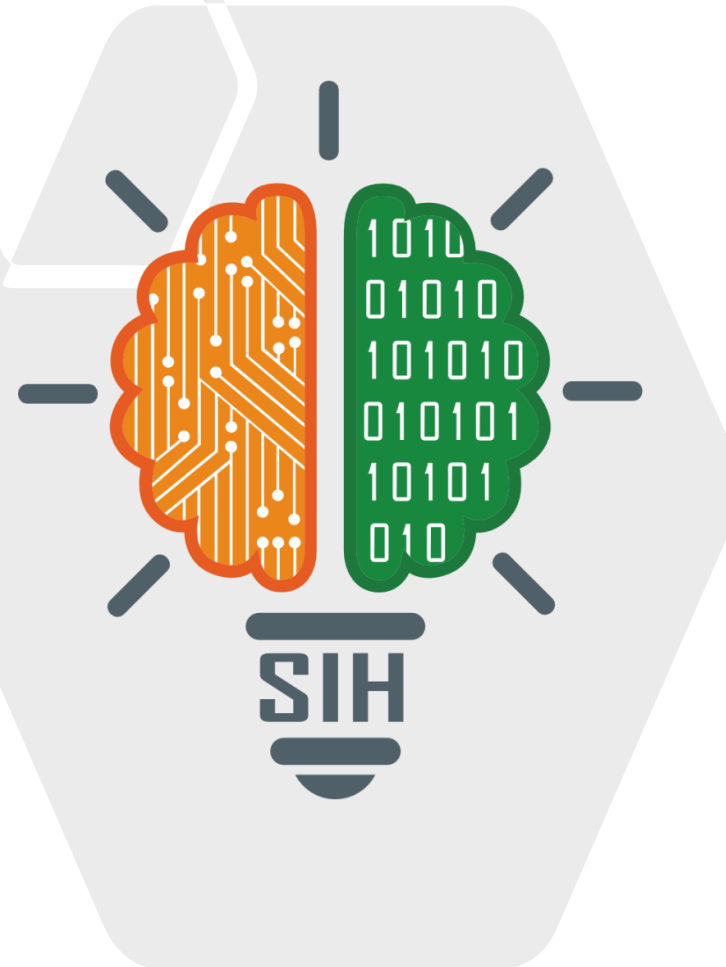
Problem Statement ID – **SIH26042**

Problem Statement Title- **AI-Powered Vernacular Pedagogy and Real-Time Translation Tool for Mother Tongue-Based Primary Education**

Theme- Smart Education

PS Category- Software Team ID-

Team Name (Registered on portal)



FROM
ZERO TO
HERO

Idea Title



Smart India Hackathon 2026

Team Name: FROM ZERO TO HERO

Problem Number : 42 (Government Of Jharkhand)

Institute: Noida Institute Of Engineering And Technology

Team Leader – Sammar Somani

Members – Rudraksha Kumar Gupta

Nitish Yadav

Taman Kumari

Shivangi Yadav

Vinayak

TAGLINE: “Learn In The Language You Think In.”

TECHNICAL APPROACH



SMART INDIA
HACKATHON
2025



**TECHNOLOGIES TO BE USED (E.G.
PROGRAMMING LANGUAGES,
FRAMEWORKS, HARDWARE)**



**METHODOLOGY AND PROCESS
FOR IMPLEMENTATION (FLOW
CHARTS/IMAGES/ WORKING
PROTOTYPE)**

PROPOSED SOLUTION



AI-Powered Vernacular Learning Assistant



“From Translation to Understanding”



Our solution bridges the language gap in primary education by combining AI translation with age-appropriate, mother-tongue pedagogy.



IMPACT AND BENEFITS



- From Teacher's Voice to Child's Understanding

- ① TEACHER INPUT 🎤
Teacher speaks in any supported language
- ② AI SPEECH RECOGNITION
Voice → Text + Language Detection
- ③ TRANSLATION ENGINE 🌐
Context-aware translation into student's mother tongue
- ④ PEDAGOGY ENGINE 🧠
Simplifies content according to Class + Age + Learning Level
- ⑤ STUDENT OUTPUT 🧒🧒
Easy-to-understand Text + Voice + Visual Explanation



RESEARCH & REFERENCES

- **1. Mother-Tongue Based Learning**
 - Children learn concepts more effectively when early education uses a language they understand. UNESCO recommends multilingual education to improve inclusion and learning.
- **2. NEP 2020 – India**
 - NEP 2020 recommends that, wherever possible, the home/mother tongue/local/familiar language should be used as the medium of instruction in the early years.
- **3. NIPUN Bharat / Foundational Learning**
 - NIPUN Bharat focuses on foundational literacy and numeracy by Grade 3. Its Foundational Learning Study covered approximately **86,000 Grade 3 students across 20 languages**.



- **4. UNESCO – Bhasha Matters 2025**

UNESCO's India 2025 report highlights mother-tongue and multilingual education as important for learning, inclusion and linguistic diversity, and recommends multilingual learning resources and language technologies.

- **5. Bhashini – Indian Language AI**

Bhashini supports technologies such as Speech Recognition, Text-to-Speech, Machine Translation and OCR for Indian languages, providing a strong technical foundation for our proposed solution.